Chapter 11. The Seifert–van Kampen Theorem

Free products and free groups, the Seifert–van Kampen theorem, wedges of circles, attaching 2-cells

Contents

11.1 Direct sums of abelian groups **2**

11.2 Free products of groups **3**

11.3 Free groups **5**

11.4 The Seifert–van Kampen theorem **6**

11.5 The fundamental group of a wedge of circles **8**

11.6 Adjoining a two-cell **9**

11.7 The fundamental groups of the torus and the dunce cap **10**

11.1 Direct sums of abelian groups

Motivation. To compute fundamental groups of spaces glued from simpler pieces we need ways to combine groups. This section recalls direct sums for abelian groups, characterized by their universal property, as preparation for the noncommutative analogue.

Abelian groups are written additively. Subgroups G_α ($\alpha \in J$) of G **generate** G if every $x \in G$ is a finite sum $\sum x_\alpha$ with $x_\alpha \in G_\alpha$ (almost all 0); G is their **direct sum**, $G = \bigoplus G_\alpha$, if moreover this expression is unique. G is **free abelian** with **basis** $\{a_\alpha\}$ if each a_α generates an infinite cyclic subgroup G_α and $G = \bigoplus G_\alpha$.

Lemma 11.1 (67.1)

If $G = \bigoplus G_\alpha$, then G has the **extension property**: for every abelian group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique homomorphism $h : G \to H$ restricting to each h_α. Conversely, if the G_α generate G and the extension property holds, then $G = \bigoplus G_\alpha$.

Proof. If $G = \bigoplus G_\alpha$, define $h(\sum x_\alpha) = \sum h_\alpha(x_\alpha)$, well defined by uniqueness and clearly a homomorphism; it is forced, so unique. Conversely suppose $\sum x_\alpha = \sum y_\alpha$. For a fixed β, let $H = G_\beta$, $h_\beta = \mathrm{id}$ and $h_\alpha = 0$ for $\alpha \neq \beta$; applying the extension h to both sides gives $x_\beta = y_\beta$. ∎

Corollary 11.2 (67.2)

If $G = G_1 \oplus G_2$, $G_1 = \bigoplus_{\alpha \in J} H_\alpha$ and $G_2 = \bigoplus_{\beta \in K} H_\beta$ with $J \cap K = \emptyset$, then $G = \bigoplus_{\gamma \in J \cup K} H_\gamma$.

Proof. The H_γ generate G. Homomorphisms $h_\gamma : H_\gamma \to H$ extend to G_1 and to G_2, and these extend to G; uniqueness follows because the H_γ generate. Apply Lemma 11.1. ∎

Corollary 11.3 (67.3)

If $G = G_1 \oplus G_2$, then $G/G_2 \cong G_1$.

Proof. Extend $\mathrm{id} : G_1 \to G_1$ and $0 : G_2 \to G_1$ to $h : G \to G_1$; it is surjective with kernel G_2. ∎

Theorem 11.4 (67.4)

For any family $\{G_\alpha\}$ of abelian groups there are an abelian group G and monomorphisms $i_\alpha : G_\alpha \to G$ with $G = \bigoplus i_\alpha(G_\alpha)$ (the **external direct sum**).

Proof. Let G be the subgroup of $\prod G_\alpha$ of tuples with only finitely many nonzero entries, and $i_\beta(x)$ the tuple with x in coordinate β and 0 elsewhere. Each element is uniquely a finite sum of such tuples. ∎

Lemma 11.5 (67.5)

If $i_\alpha : G_\alpha \to G$ are monomorphisms and $G = \bigoplus i_\alpha(G_\alpha)$, then for any abelian H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique $h : G \to H$ with $h \circ i_\alpha = h_\alpha$.

Proof. Apply Lemma 11.1 to the homomorphisms $h_\alpha \circ i_\alpha^{-1} : i_\alpha(G_\alpha) \to H$. ∎

Theorem 11.6 (67.6, uniqueness of direct sums)

If $G = \bigoplus i_\alpha(G_\alpha)$ and $G' = \bigoplus i'_\alpha(G_\alpha)$ with monomorphisms i_α, i'_α, there is a unique isomorphism $\phi : G \to G'$ with $\phi \circ i_\alpha = i'_\alpha$.

Proof. Lemma 11.5 gives $\phi : G \to G'$ with $\phi i_\alpha = i'_\alpha$ and $\psi : G' \to G$ with $\psi i'_\alpha = i_\alpha$. Then $\psi\phi i_\alpha = i_\alpha$, and by the uniqueness in Lemma 11.5 (with $H = G$, $h_\alpha = i_\alpha$), $\psi\phi = \mathrm{id}$; similarly $\phi\psi = \mathrm{id}$. ■

Lemma 11.7 (67.7)

Let $\{a_\alpha\}$ generate the abelian group G. Then G is free abelian with basis $\{a_\alpha\}$ iff for every abelian H and elements $y_\alpha \in H$ there is a homomorphism $h : G \to H$ with $h(a_\alpha) = y_\alpha$; such h is then unique.

Proof. Let G_α be the subgroup generated by a_α. If the condition holds, each a_α has infinite order: with $H = \mathbb{Z}$ and all $y_\alpha = 1$, $na_\alpha = 0$ forces $n = 0$. Homomorphisms $G_\alpha \to H$ correspond to choices of y_α, so the extension property of Lemma 11.1 holds, and $G = \bigoplus G_\alpha$. Conversely, if G is free abelian, $na_\alpha \mapsto ny_\alpha$ defines h_α on the infinite cyclic G_α; extend by Lemma 11.1. Uniqueness holds because the a_α generate. ■

Theorem 11.8 (67.8)

If G is free abelian with basis $\{a_1, \dots, a_n\}$, then n is determined by G.

Proof. $G \cong \mathbb{Z}^n$, and $2G = \{2x\}$ consists of the elements $\sum n_i a_i$ with all n_i even. The map $\sum n_i a_i \mapsto (n_i \bmod 2)$ is a surjection $G \to (\mathbb{Z}/2)^n$ with kernel $2G$, so $G/2G$ has 2^n elements. The number $|G/2G|$ depends only on G. ■

n is the **rank** of G.

11.2 Free products of groups

Motivation. The fundamental group of a figure eight is not abelian: going around one loop and then the other differs from the reverse. The free product is the group-theoretic way of combining two groups with no relations between them, and it is what van Kampen's theorem produces.

Let G be a group and $\{G_\alpha\}$ subgroups generating G with $G_\alpha \cap G_\beta = \{1\}$ for $\alpha \neq \beta$. A **word** is a finite sequence $(g_1, \dots, g_n)$ of elements of $\bigcup G_\alpha$, representing the product $g_1 \cdots g_n$; it is **reduced** if no $g_i = 1$ and no two consecutive g_i, g_{i+1} lie in the same G_α. Every element of G is represented by a reduced word (combine adjacent letters in one group and delete 1's), the identity by the empty word. G is the **free product** of the G_α, $G = \prod^* G_\alpha$, if each element is represented by only one reduced word. It suffices that no nonempty reduced word represent 1: if reduced words $(g_1, \dots, g_n)$ and $(h_1, \dots, h_m)$ represent the same element, then $g_1 \cdots g_n h_m^{-1} \cdots h_1^{-1} = 1$; if g_n, h_m lie in different groups this word is reduced and nonempty, if they lie in one group and differ, merging them gives a nonempty reduced word, and if $g_n = h_m$ we cancel and induct.

Lemma 11.9 (68.1)

If $G = \prod^* G_\alpha$, then for every group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique homomorphism $h : G \to H$ restricting to each h_α.

Proof. For x represented by the reduced word $(g_1, \dots, g_n)$ with $g_i \in G_{\alpha_i}$, set $h(x) = h_{\alpha_1}(g_1) \cdots h_{\alpha_n}(g_n)$; this is well defined by uniqueness. The same formula, applied to any word, gives the same value, since combining adjacent letters of one group or deleting 1's does not change the product of the images. The concatenation of words representing x and y represents xy, so $h(xy) = h(x)h(y)$. Uniqueness holds because the G_α generate G. ∎

Theorem 11.10 (68.2)

For any family $\{G_\alpha\}$ of groups there are a group G and monomorphisms $i_\alpha : G_\alpha \to G$ such that G is the free product of the subgroups $i_\alpha(G_\alpha)$ (the **external free product**).

Proof. We may assume the sets $G_\alpha - \{1\}$ disjoint. Let W be the set of reduced words in their union (including the empty word). For $g \in G_\alpha$ define $\pi_g : W \to W$: $\pi_1 = \mathrm{id}$; for $g \neq 1$ and $w = (g_1, \dots, g_n)$, $\pi_g(w) = (g, g_1, \dots, g_n)$ if w is empty or $g_1 \notin G_\alpha$; $(gg_1, g_2, \dots, g_n)$ if $g_1 \in G_\alpha$ and $gg_1 \neq 1$; and $(g_2, \dots, g_n)$ if $gg_1 = 1$. Checking the cases shows $\pi_g \pi_{g'} = \pi_{gg'}$ for $g, g' \in G_\alpha$ (both sides only modify the first letter when it lies in G_α). Hence each π_g is a bijection with inverse $\pi_{g^{-1}}$, and $i_\alpha(g) = \pi_g$ is a homomorphism into the group of permutations of W; it is injective because $\pi_g(\emptyset) = (g)$. Let G be the group generated by all π_g. If $(g_1, \dots, g_n)$ is a nonempty reduced word, then $\pi_{g_1} \cdots \pi_{g_n}(\emptyset) = (g_1, \dots, g_n) \neq \emptyset$, so it does not represent the identity. Also $i_\alpha(G_\alpha) \cap i_\beta(G_\beta) = \{1\}$ for the same reason. ∎

Lemma 11.11 (68.3)

If $i_\alpha : G_\alpha \to G$ are monomorphisms and G is the free product of the $i_\alpha(G_\alpha)$, then for every group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique $h : G \to H$ with $h \circ i_\alpha = h_\alpha$.

Proof. Apply Lemma 11.9 to $h_\alpha \circ i_\alpha^{-1}$. ∎

Theorem 11.12 (68.4, uniqueness of free products)

Let $i_\alpha : G_\alpha \to G$ and $i'_\alpha : G_\alpha \to G'$ be monomorphisms whose images generate G and G', both with the extension property of Lemma 11.11. Then there is a unique isomorphism $\phi : G \to G'$ with $\phi \circ i_\alpha = i'_\alpha$.

Proof. As in Theorem 11.6: the extensions ϕ, ψ satisfy $\psi\phi i_\alpha = i_\alpha$, so $\psi\phi = \mathrm{id}$ by uniqueness, and likewise $\phi\psi = \mathrm{id}$. ∎

Lemma 11.13 (68.5)

If homomorphisms $i_\alpha : G_\alpha \to G$ satisfy the extension condition of Lemma 11.11, then each i_α is a monomorphism and G is the free product of the $i_\alpha(G_\alpha)$.

Proof. For a fixed β, take $H = G_\beta$, $h_\beta = \mathrm{id}$ and h_α trivial for $\alpha \neq \beta$; the extension h has $h \circ i_\beta = \mathrm{id}$, so i_β is injective. Let G' with i'_α be the external free product (Theorem 11.10), which has the extension property. Extensions give $\phi : G \to G'$ with $\phi i_\alpha = i'_\alpha$ and $\psi : G' \to G$ with $\psi i'_\alpha = i_\alpha$. By the uniqueness in the extension condition for G, $\psi\phi = \mathrm{id}_G$, and since G' is generated by the $i'_\alpha(G_\alpha)$, $\phi\psi = \mathrm{id}$. So ϕ is an isomorphism carrying $i_\alpha(G_\alpha)$ onto $i'_\alpha(G_\alpha)$, and G is the free product of the $i_\alpha(G_\alpha)$. ∎

Corollary 11.14 (68.6)

If $G = G_1 * G_2$ with $G_1 = \prod^*_{\alpha \in J} H_\alpha$, $G_2 = \prod^*_{\beta \in K} H_\beta$ and $J \cap K = \emptyset$, then $G = \prod^*_{\gamma \in J \cup K} H_\gamma$.

Proof. Homomorphisms $h_\gamma : H_\gamma \to H$ extend to G_1 and G_2, and then to G; the extension is unique since the H_γ generate G. Apply Lemma 11.13 to the inclusions. ■

Theorem 11.15 (68.7)

Let $G = G_1 * G_2$, $N_i \lhd G_i$, and N the least normal subgroup of G containing N_1 and N_2. Then $G/N \cong (G_1/N_1) * (G_2/N_2)$.

Proof. Let $F = (G_1/N_1) * (G_2/N_2)$. The projections $G_i \to G_i/N_i \to F$ extend to $\phi : G \to F$, which is surjective and kills N_1, N_2, hence N; it induces $\bar{\phi} : G/N \to F$. The maps $G_i/N_i \to G/N$, $gN_i \mapsto gN$, are well-defined homomorphisms ($N_i \subset N$) and extend to $\psi : F \to G/N$. On generators gN ($g \in G_i$) and gN_i, $\psi\bar{\phi}$ and $\bar{\phi}\psi$ are the identity; hence they are the identity. ■

Corollary 11.16 (68.8)

If N is the least normal subgroup of $G_1 * G_2$ containing G_1, then $(G_1 * G_2)/N \cong G_2$.

Proof. Take $N_1 = G_1$ and $N_2 = \{1\}$ in Theorem 11.15. ■

Lemma 11.17 (68.9)

The least normal subgroup N of G containing $S \subset G$ is generated by the conjugates of elements of S.

Proof. The subgroup N' generated by these conjugates contains S and is normal, since $g(x_1 \cdots x_k)g^{-1} = (gx_1g^{-1}) \cdots (gx_kg^{-1})$ and a conjugate of a conjugate is a conjugate. Every normal subgroup containing S contains N', so $N = N'$. ■

11.3 Free groups

Motivation. Free groups are free products of copies of $\mathbb{Z}$. They are the fundamental groups of graphs, and every group is a quotient of a free group, which is how presentations of fundamental groups arise.

G is a **free group** with **system of free generators** $\{a_\alpha\}$ if each a_α generates an infinite cyclic subgroup G_α and $G = \prod^* G_\alpha$.

Lemma 11.18 (69.1)

If G is free on $\{a_\alpha\}$, then for every group H and elements $y_\alpha \in H$ there is a unique homomorphism $h : G \to H$ with $h(a_\alpha) = y_\alpha$. Conversely, if the a_α generate G and this extension condition holds, G is free on $\{a_\alpha\}$.

Proof. If G is free, $a_\alpha^n \mapsto y_\alpha^n$ defines homomorphisms on the infinite cyclic G_α; extend by Lemma 11.9. Conversely, with $H = \mathbb{Z}$ and $y_\alpha = 1$, $a_\alpha^n = 1$ forces $n = 0$, so G_α is infinite cyclic; homomorphisms $G_\alpha \to H$ correspond to the choice of y_α, so the extension condition of Lemma 11.11 holds for the inclusions, and Lemma 11.13 applies. ■

Theorem 11.19 (69.2)

If $G = G_1 * G_2$ with G_1, G_2 free on $\{a_\alpha\}_{\alpha\in J}$ and $\{a_\alpha\}_{\alpha\in K}$, $J \cap K = \emptyset$, then G is free on $\{a_\alpha\}_{\alpha\in J\cup K}$.

Proof. Corollary 11.14. ■

If G is free on $\{a_\alpha\}$ and $\{r_\beta\} \subset G$, and N is the least normal subgroup containing the r_β, a group isomorphic to G/N has the **presentation** with generators a_α and relations r_β. The **commutator subgroup** $[G, G]$ is generated by all commutators $[x, y] = xyx^{-1}y^{-1}$.

Lemma 11.20 (69.3)

$[G, G]$ is normal and $G/[G, G]$ is abelian. Every homomorphism $h : G \to H$ into an abelian group has $[G, G] \subset \ker h$, so it induces $k : G/[G, G] \to H$.

Proof. $g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}]$, so conjugates of products of commutators are products of commutators. $xy(yx)^{-1} = [x, y]$, so cosets of x and y commute. If H is abelian, $h([x, y]) = 1$. ■

Theorem 11.21 (69.4)

If G is free on $\{a_\alpha\}$, then $G/[G, G]$ is free abelian with basis the cosets $[a_\alpha]$.

Proof. The cosets generate. Given y_α in an abelian H, Lemma 11.18 gives h with $h(a_\alpha) = y_\alpha$, which induces k with $k([a_\alpha]) = y_\alpha$ (Lemma 11.20). Apply Lemma 11.7. ■

Corollary 11.22 (69.5)

If G is free on n generators, every system of free generators of G has n elements.

Proof. $G/[G, G]$ is free abelian of rank equal to the number of free generators (Theorem 11.21), and the rank is unique (Theorem 11.8). (An infinite system cannot occur: $G/[G, G]$ would then not be finitely generated.) ■

11.4 The Seifert–van Kampen theorem

Motivation. This is the main computational tool for fundamental groups: if $X = U \cup V$ with $U, V, U \cap V$ path connected, then $\pi_1(X)$ is the free product of $\pi_1(U)$ and $\pi_1(V)$ with the images of $\pi_1(U \cap V)$ identified.

Theorem 11.23 (70.1, Seifert–van Kampen theorem)

Let $X = U \cup V$ with U, V open and U, V, $U \cap V$ path connected, $x_0 \in U \cap V$. Let i_1, i_2 be induced by the inclusions of $U \cap V$ into U and V, and j_1, j_2 by the inclusions of U and V into X. If $\phi_1 : \pi_1(U, x_0) \to H$ and $\phi_2 : \pi_1(V, x_0) \to H$ are homomorphisms with $\phi_1 \circ i_1 = \phi_2 \circ i_2$, there is a unique homomorphism $\Phi : \pi_1(X, x_0) \to H$ with $\Phi \circ j_1 = \phi_1$ and $\Phi \circ j_2 = \phi_2$.

Proof. Uniqueness: $\pi_1(X, x_0)$ is generated by the images of j_1, j_2 (Theorem 9.38). Write $[f]_U, [f]_V$ for path classes in U, V and $[f]$ for classes in X.

Step 1. for a loop f at x_0 in U or V, let $\rho(f) = \phi_1([f]_U)$ or $\phi_2([f]_V)$; if f lies in $U \cap V$ the two agree since $\phi_1 i_1 = \phi_2 i_2$. Then (1) $[f]_U = [g]_U$ or $[f]_V = [g]_V$ implies $\rho(f) = \rho(g)$;

(2) if f, g both lie in U or both in V, $\rho(f * g) = \rho(f)\rho(g)$.

Step 2. for each $x \in X$ choose a path α_x from x_0 to x: constant if $x = x_0$, in $U \cap V$ if $x \in U \cap V$, and in U (resp. V) if x lies only in U (resp. V). For a path f in U or V from x to y, $L(f) = \alpha_x * f * \bar{\alpha}_y$ is a loop at x_0 in U or V; let $\sigma(f) = \rho(L(f))$. Then σ extends ρ (for a loop at x_0, $L(f) \simeq_p f$ in U or V), satisfies (1), and satisfies (2) whenever $f * g$ is defined, since $L(f) * L(g) \simeq_p L(f * g)$ in U or V.

Step 3. for an arbitrary path f in X, choose $0 = s_0 < \cdots < s_n = 1$ with each $[s_{i-1}, s_i]$ mapped into U or V, let f_i be the reparametrized pieces, and set $\tau(f) = \sigma(f_1) \cdots \sigma(f_n)$. Adding a point p to the subdivision replaces $\sigma(f_i)$ by $\sigma(f_i')\sigma(f_i'')$, which is equal since $f_i \simeq_p f_i' * f_i''$ in U or V; so τ is well defined (any two subdivisions have a common refinement) and extends σ.

Step 4. if $f \simeq_p g$ in X, then $\tau(f) = \tau(g)$. Special case: a path homotopy F for which some subdivision $s_0 < \cdots < s_n$ has each $R_i = [s_{i-1}, s_i] \times I$ mapped into U or V. Let $\beta_i(t) = F(s_i, t)$, so β_0, β_n are constant. In R_i the path along the bottom and right edges and the path along the left and top edges are path homotopic (convexity); applying F gives $f_i * \beta_i \simeq_p \beta_{i-1} * g_i$ in U or V, so $\sigma(f_i)\sigma(\beta_i) = \sigma(\beta_{i-1})\sigma(g_i)$. Also $\sigma(\beta_0) = \sigma(\beta_n) = 1$ (from $\beta * \beta = \beta$ up to path homotopy for a constant path). Multiplying, $\tau(f) = \sigma(f_1) \cdots \sigma(f_n) = \sigma(g_1) \cdots \sigma(g_n) = \tau(g)$. General case: subdivide $I \times I$ into rectangles $[s_{i-1}, s_i] \times [t_{j-1}, t_j]$ mapped into U or V and apply the special case to consecutive paths $f_j(s) = F(s, t_j)$.

Step 5. $\tau(f * g) = \tau(f)\tau(g)$: use a subdivision of I containing $\frac{1}{2}$ for $f * g$; its two halves give subdivisions for f and g, and the factors match.

Step 6. define $\Phi([f]) = \tau(f)$. By Steps 4 and 5 it is a well-defined homomorphism, and for a loop f in U, $\Phi(j_1[f]_U) = \tau(f) = \rho(f) = \phi_1([f]_U)$; similarly for V. ■

Theorem 11.24 (70.2, Seifert–van Kampen theorem, classical version)

Under the same hypotheses, let $j : \pi_1(U, x_0) * \pi_1(V, x_0) \to \pi_1(X, x_0)$ extend j_1 and j_2. Then j is surjective, and its kernel is the least normal subgroup N containing all elements $i_1(g)^{-1} i_2(g)$, $g \in \pi_1(U \cap V, x_0)$.

Proof. Surjectivity is Theorem 9.38. Since $j_1 i_1 = j_2 i_2$ (both are induced by the inclusion of $U \cap V$ into X), each $i_1(g)^{-1} i_2(g)$ lies in $\ker j$, so $N \subset \ker j$ and j induces an epimorphism $k : F/N \to \pi_1(X, x_0)$, F the free product. Let $H = F/N$ and ϕ_1, ϕ_2 the inclusions into F followed by projection. Then $\phi_1 i_1(g) = i_1(g)N = i_2(g)N = \phi_2 i_2(g)$, so Theorem 11.23 gives Φ with $\Phi j_1 = \phi_1$, $\Phi j_2 = \phi_2$. For $g \in \pi_1(U, x_0)$, $\Phi k(gN) = \Phi j_1(g) = gN$, and similarly for V; these cosets generate H, so $\Phi \circ k = \mathrm{id}$ and k is injective. Hence $\ker j = N$. ■

Corollary 11.25 (70.3)

If moreover $U \cap V$ is simply connected, then $\pi_1(U, x_0) * \pi_1(V, x_0) \cong \pi_1(X, x_0)$.

Corollary 11.26 (70.4)

If V is simply connected, then $\pi_1(U, x_0)/N \cong \pi_1(X, x_0)$, where N is the least normal subgroup of $\pi_1(U, x_0)$ containing the image of i_1.

Proof. In Theorem 11.24, N is trivial in the first case; in the second, the free product is $\pi_1(U, x_0)$ and $i_1(g)^{-1} i_2(g) = i_1(g)^{-1}$. ■

Example 11.1

A theta space $X = A \cup B \cup C$ (arcs meeting in their end points) has free fundamental group of rank 2: with a, b interior points of A, B, let $U = X - a$, $V = X - b$. $U \cap V$ is contractible, while U and V deformation retract onto the circles $B \cup C$ and $A \cup C$. Apply Corollary 11.25.

11.5 The fundamental group of a wedge of circles

Motivation. The first application: a wedge of n circles has fundamental group free on n generators.

A Hausdorff space X that is the union of subspaces $S_1, \ldots, S_n$, each homeomorphic to S^1, with $S_i \cap S_j = \{p\}$ for $i \neq j$, is the **wedge** of these circles.

Theorem 11.27 (71.1)

If X is the wedge of $S_1, \ldots, S_n$ and f_i is a loop in S_i representing a generator of $\pi_1(S_i, p)$, then $\pi_1(X, p)$ is free with free generators $[f_1], \ldots, [f_n]$.

Proof. Induction on n, the case $n = 1$ being clear. Choose $q_i \in S_i - p$ and let $W_i = S_i - q_i$, $U = S_1 \cup W_2 \cup \cdots \cup W_n$, $V = W_1 \cup S_2 \cup \cdots \cup S_n$. Then $U \cap V = W_1 \cup \cdots \cup W_n$; U, V, $U \cap V$ are open and path connected. Each W_i is an open arc deformation retracting onto p; these deformations fit together, by the pasting lemma on the closed sets $W_i \times I$ of $(U \cap V) \times I$, to a deformation retraction of $U \cap V$ onto p. So $U \cap V$ is simply connected and $\pi_1(X, p) \cong \pi_1(U, p) * \pi_1(V, p)$ by Corollary 11.25. Similarly U deformation retracts onto S_1 and V onto $S_2 \cup \cdots \cup S_n$; so $\pi_1(U, p)$ is infinite cyclic generated by $[f_1]$ and, by induction, $\pi_1(V, p)$ is free on $[f_2], \ldots, [f_n]$. Apply Theorem 11.19. ■

A space X that is the union of subspaces X_α has the topology **coherent** with them if $C \subset X$ is closed whenever each $C \cap X_\alpha$ is closed in X_α. If X is the union of subspaces S_α ($\alpha \in J$) homeomorphic to S^1 with $S_\alpha \cap S_\beta = \{p\}$ for $\alpha \neq \beta$, and the topology is coherent with them, X is the **wedge** of the circles S_α (for finite J this agrees with the earlier definition, a finite union of closed sets being coherent).

Lemma 11.28 (71.2)

A wedge X of circles S_α is normal, and every compact subspace lies in the union of finitely many S_α.

Proof. Points are closed by coherence. Let A, B be disjoint closed with $p \notin B$. In each S_α choose disjoint open sets $U_\alpha \supset \{p\} \cup (A \cap S_\alpha)$ and $V_\alpha \supset B \cap S_\alpha$ (S_α is normal). $U = \bigcup U_\alpha$ and $V = \bigcup V_\alpha$ are disjoint, and $U \cap S_\alpha = U_\alpha$ (all U_β contain p, the only common point), $V \cap S_\alpha = V_\alpha$ (no V_β contains p); by coherence U, V are open. If C is compact, choose $x_\alpha \in C \cap (S_\alpha - p)$ whenever possible. Every subset of $D = \{x_\alpha\}$ meets each S_α in at most one point, hence is closed; so D is closed and discrete in C, and as C is limit point compact, D is finite. ■

Theorem 11.29 (71.3)

If X is the wedge of circles S_α and f_α represents a generator of $\pi_1(S_\alpha, p)$, then $\pi_1(X, p)$ is free with free generators $\{[f_\alpha]\}$.

Proof. Let G_α be the image of $\pi_1(S_\alpha, p)$. A loop or a path homotopy in X has compact image, so lies in a finite union of circles (Lemma 11.28). Hence: the G_α generate $\pi_1(X, p)$ (by Theorem 11.27 in a finite wedge); each $\pi_1(S_\alpha, p) \to G_\alpha$ is injective, so G_α is infinite cyclic; and a nonempty reduced

word in the G_α representing the identity would give a loop nulhomotopic in some finite wedge, contradicting Theorem 11.27. ■

Example 11.2

The **infinite earring**, the union $X \subset \mathbb{R}^2$ of the circles C_n of radius $1/n$ centred at $(1/n, 0)$, is not a wedge: its intersection with the positive x-axis meets each C_n in one point but is not closed. Indeed the classes $[f_n]$ do not generate $\pi_1(X, p)$: let g run through f_n on $[1/(n+1), 1/n]$, with $g(0) = p$, a continuous loop. For $N > n$, the retraction of X onto C_N (collapsing the other circles to p) kills $[f_1], \ldots, [f_n]$ but sends $[g]$ to the generator $[f_N]$; so $[g]$ is not in the subgroup generated by $[f_1], \ldots, [f_n]$ for any n.

Lemma 11.30 (71.4)

For every index set J there is a wedge of circles S_α, $\alpha \in J$.

Proof. Give J the discrete topology, let $E = S^1 \times J$, and let X be the quotient of E collapsing $P = b_0 \times J$ to a point p, with quotient map π; put $S_\alpha = \pi(S^1 \times \alpha)$. If C is closed in $S^1 \times \alpha$, then $\pi^{-1}\pi(C)$ is C or $C \cup P$, closed, so $\pi(C)$ is closed. Hence S_α is closed and π maps $S^1 \times \alpha$ homeomorphically onto S_α. If $D \cap S_\alpha$ is closed for each α, then $\pi^{-1}(D) \cap (S^1 \times \alpha)$ is closed for each α, so $\pi^{-1}(D)$ is closed and D is closed: the topology is coherent. ■

11.6 Adjoining a two-cell

Motivation. Attaching a disc along a loop kills that loop. In group terms, adjoining a 2-cell adds one relation, so cell structures translate directly into presentations of π_1.

Theorem 11.31 (72.1)

Let X be Hausdorff, $A \subset X$ closed and path connected, and $h : B^2 \to X$ continuous, mapping $\operatorname{Int} B^2$ bijectively onto $X - A$ and S^1 into A. Let $p \in S^1$, $a = h(p)$ and $k = h|S^1 : (S^1, p) \to (A, a)$. Then $i_* : \pi_1(A, a) \to \pi_1(X, a)$ induced by inclusion is surjective, and its kernel is the least normal subgroup containing $k_*(\pi_1(S^1, p))$.

Proof.

Step 1. with $x_0 = h(0)$ and $U = X - x_0$, A is a deformation retract of U. Let $C = h(B^2)$, closed in X, and $\pi : B^2 \to C$ the restriction of h; $\pi \times \mathrm{id} : B^2 \times I \to C \times I$ is closed (compact to Hausdorff), hence a quotient map, and so is its restriction to the saturated open set $(B^2 - 0) \times I$. The radial deformation retraction of $B^2 - 0$ onto S^1 (which fixes S^1) therefore induces a deformation retraction of $C - x_0$ onto $\pi(S^1)$; extend it by keeping A fixed. By the pasting lemma (the closed sets $C - x_0$ and A of U overlap in $\pi(S^1)$, which stays fixed) this is a deformation retraction of U onto A. So $\pi_1(A, a) \to \pi_1(U, a)$ is an isomorphism (Theorem 9.33), and it suffices to show: $\pi_1(U, a) \to \pi_1(X, a)$ is surjective with kernel the least normal subgroup containing $[g]$, $g = h \circ f$, f a generator of $\pi_1(S^1, p)$.

Step 2. let $V = X - A = h(\operatorname{Int} B^2)$, open. $\pi|\operatorname{Int} B^2$ is a quotient map (restriction to a saturated open set) and bijective, so a homeomorphism; V is simply connected. $U \cap V = V - x_0 \cong \operatorname{Int} B^2 - 0$ is path connected with infinite cyclic fundamental group, and U is path connected. For $b \in U \cap V$, Corollary 11.26 shows $\pi_1(U, b) \to \pi_1(X, b)$ is surjective with kernel the least normal subgroup containing the image of $\pi_1(U \cap V, b)$.

Step 3. let q be the midpoint of the segment from 0 to p, $b = h(q)$, f_0 a loop in $\operatorname{Int} B^2 - 0$ at q generating its fundamental group, $g_0 = h \circ f_0$, γ the segment from q to p and $\delta = h \circ \gamma$. The isomorphisms $\hat{\delta}$ commute with the homomorphisms induced by inclusion $U \to X$, so $\pi_1(U, a) \to \pi_1(X, a)$ is surjective with kernel the least normal subgroup containing $\hat{\delta}([g_0])$. Now $\bar{\gamma} * f_0 * \gamma$ generates $\pi_1(B^2 - 0, p)$, so it is path homotopic in $B^2 - 0$ to f or $\bar{f}$; applying h, $\hat{\delta}([g_0]) = [g]^{\pm 1}$ in $\pi_1(U, a)$. ■

A space homeomorphic to B^2 is a **2-cell**; X is obtained from A by **adjoining a 2-cell**.

11.7 The fundamental groups of the torus and the dunce cap

Motivation. Applying the two-cell result to standard polygon models gives presentations for the torus, $\langle a, b \mid aba^{-1}b^{-1}\rangle \cong \mathbb{Z}^2$, and for the dunce cap.

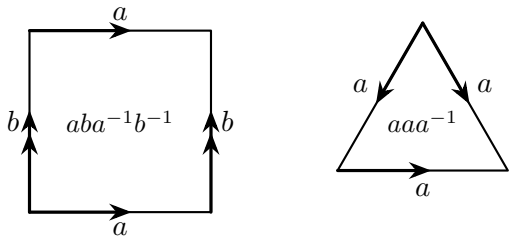


Figure. *Left: the torus is a square with boundary word $aba^{-1}b^{-1}$, so $\pi_1 = \langle a, b \mid aba^{-1}b^{-1}\rangle \cong \mathbb{Z} \times \mathbb{Z}$. Right: a dunce cap is a triangle with all three edges identified as a, a, a^{-1} (reading around the boundary), so $\pi_1 = \langle a \mid a^2a^{-1}\rangle$ is trivial.*

Theorem 11.32 (73.1)

The fundamental group of the torus has a presentation with two generators α, β and one relation $\alpha\beta\alpha^{-1}\beta^{-1}$.

Proof. Let $h : I^2 \to T = S^1 \times S^1$ be the restriction of $p \times p$, $A = h(\operatorname{Bd} I^2)$, a wedge of two circles, and $a = h(0, 0)$; the hypotheses of Theorem 11.31 hold ($I^2 \cong B^2$). With $a_0(t) = (t, 0)$, $b_0(t) = (0, t)$, $a_1(t) = (t, 1)$, $b_1(t) = (1, t)$, the loops $\alpha = h \circ a_0$, $\beta = h \circ b_0$ are free generators of $\pi_1(A, a)$ (Theorem 11.27). The loop $f = a_0 * (b_1 * (\bar{a}_1 * \bar{b}_0))$ generates $\pi_1(\operatorname{Bd} I^2, p)$, and $h \circ f = \alpha * (\beta * (\bar{\alpha} * \bar{\beta}))$. Apply Theorem 11.31. ■

Corollary 11.33 (73.2)

The fundamental group of the torus is free abelian of rank 2.

Proof. Let G be free on α, β and N the least normal subgroup containing $\alpha\beta\alpha^{-1}\beta^{-1}$. It is a commutator, so $N \subset [G, G]$; and G/N is abelian, being generated by the commuting cosets $\alpha N, \beta N$, so $[G, G] \subset N$. Hence $G/N = G/[G, G]$, free abelian of rank 2 (Theorem 11.21). ■

For $n > 1$ let r be rotation of S^1 by $2\pi/n$. The **n-fold dunce cap** X is the quotient of B^2 identifying each $x \in S^1$ with $r(x), \ldots, r^{n-1}(x)$. The quotient map π is closed: for closed C, $\pi^{-1}\pi(C) = C \cup r(C_0) \cup \cdots \cup r^{n-1}(C_0)$ with $C_0 = C \cap S^1$. So X is compact, and Hausdorff by the following lemma.

Lemma 11.34 (73.3)

If $\pi : E \to X$ is a closed quotient map and E is normal, so is X.

Proof. Points of X are closed since points of E are. For disjoint closed $A, B \subset X$, take disjoint open $U \supset \pi^{-1}(A)$, $V \supset \pi^{-1}(B)$. The sets $U_0 = X - \pi(E - U)$ and $V_0 = X - \pi(E - V)$ are open (as π is closed), contain A and B, and are disjoint: if $x \in U_0$, then $\pi^{-1}(x) \subset U$, and if $x \in V_0$, $\pi^{-1}(x) \subset V$. ■

The 2-fold dunce cap is homeomorphic to P^2: following the homeomorphism of B^2 onto the upper hemisphere by $S^2 \to P^2$ gives a closed continuous surjection that is injective on Int B^2 and identifies x with $-x$ on S^1.

Theorem 11.35 (73.4)

The fundamental group of the n-fold dunce cap is cyclic of order n.

Proof. Let $h : B^2 \to X$ be the quotient map, $A = h(S^1)$, $p = (1, 0)$, $a = h(p)$. h maps the arc of S^1 from p to $r(p)$ onto A, identifying only its end points, so A is a simple closed curve and, with $\gamma(t) = (\cos 2\pi t/n, \sin 2\pi t/n)$, $\alpha = h \circ \gamma$ generates $\pi_1(A, a) \cong \mathbb{Z}$. The loop $f = \gamma * (r \circ \gamma) * \cdots * (r^{n-1} \circ \gamma)$ generates $\pi_1(S^1, p)$, and since $h \circ r^m = h$ on S^1, $h \circ f$ is the n-fold product of α. By Theorem 11.31, $\pi_1(X, a) \cong \mathbb{Z}/n\mathbb{Z}$. ■